MySQL 8.0 Command Line Cli

```
| SalespersonID | SalespersonName | City   |
+---------------+-----------------+--------+
|             1 | Amit            | Mumbai |
|             3 | Rahul           | Mumbai |
|             5 | Neha            | Mumbai |
+---------------+-----------------+--------+
3 rows in set (0.00 sec)

mysql> SELECT MAX(Salary) AS HighestSalary
    -> FROM Emp;
+---------------+
| HighestSalary |
+---------------+
|      88000.00 |
+---------------+
1 row in set (0.00 sec)

mysql> SELECT MAX(Salary) AS SecondHighestSalary
    -> FROM Emp
    -> WHERE Salary < (
    ->     SELECT MAX(Salary)
    ->     FROM Emp
    -> );
+---------------------+
| SecondHighestSalary |
+---------------------+
|            82500.00 |
+---------------------+
1 row in set (0.01 sec)

mysql> SELECT *
    -> FROM Emp
    -> ORDER BY JoiningDate DESC;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |
+-------+---------+----------+--------+-------------+
7 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> CREATE VIEW Mumbai_Salespeople AS
    -> SELECT *
    -> FROM Salesperson
    -> WHERE City = 'Mumbai';
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Mumbai_Salespeople;
+---------------+-----------------+--------+
| SalespersonID | SalespersonName | City   |
+---------------+-----------------+--------+
|             1 | Amit            | Mumbai |
|             3 | Rahul           | Mumbai |
|             5 | Neha            | Mumbai |
+---------------+-----------------+--------+
3 rows in set (0.00 sec)

mysql> CREATE VIEW Mumbai_Salespeople AS
    -> SELECT *
    -> FROM Salesperson
    -> WHERE City = 'Mumbai';
ERROR 1050 (42S01): Table 'Mumbai_Salespeople' already exists
mysql> SELECT *
    -> FROM Mumbai_Salespeople;
+---------------+-----------------+--------+
| SalespersonID | SalespersonName | City   |
+---------------+-----------------+--------+
|             1 | Amit            | Mumbai |
|             3 | Rahul           | Mumbai |
|             5 | Neha            | Mumbai |
+---------------+-----------------+--------+
3 rows in set (0.00 sec)

mysql> SELECT MAX(Salary) AS HighestSalary
    -> FROM Emp;
+---------------+
| HighestSalary |
+---------------+
|      88000.00 |
+---------------+
1 row in set (0.00 sec)

mysql> SELECT MAX(Salary) AS SecondHighestSalary
    -> FROM Emp
    -> WHERE Salary < (
    ->     SELECT MAX(Salary)
    ->     FROM Emp
```

MySQL 8.0 Command Line Cli

```
    -> FROM Department d
    ->
    -> LEFT JOIN Emp e ON d.DeptID = e.DeptID
    ->
    -> GROUP BY d.DeptID, d.Dept_name;
+-----------+----------------+
| Dept_name | TotalEmployees |
+-----------+----------------+
| IT        |              3 |
| HR        |              2 |
| Sales     |              2 |
| Finance   |              0 |
+-----------+----------------+
4 rows in set (0.00 sec)

mysql> SELECT d.DeptID, d.Dept_name
    ->
    -> FROM Department d
    ->
    -> LEFT JOIN Emp e ON d.DeptID = e.DeptID
    ->
    -> WHERE e.EmpID IS NULL;
+--------+-----------+
| DeptID | Dept_name |
+--------+-----------+
|     40 | Finance   |
+--------+-----------+
1 row in set (0.00 sec)

mysql> CREATE VIEW Mumbai_Salespeople AS
    -> SELECT *
    -> FROM Salesperson
    -> WHERE City = 'Mumbai';
ERROR 1146 (42S02): Table 'employee.salesperson' doesn't exist
mysql> CREATE TABLE Salesperson (
    ->     SalespersonID INT PRIMARY KEY,
    ->     SalespersonName VARCHAR(50),
    ->     City VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Salesperson (SalespersonID, SalespersonName, City)
    -> VALUES
    -> (1, 'Amit', 'Mumbai'),
    -> (2, 'Priya', 'Pune'),
    -> (3, 'Rahul', 'Mumbai'),
    -> (4, 'Sneha', 'Delhi'),
```

MySQL 8.0 Command Line Cli

```
+-----------+----------------+
4 rows in set (0.00 sec)

mysql> SELECT d.DepartmentName,
    ->
    ->        COUNT(e.EmpID) AS TotalEmployees
    ->
    -> FROM Department d
    ->
    -> LEFT JOIN Emp e
    ->
    -> ON d.DeptID = e.DeptID
    ->
    -> GROUP BY d.DeptID, d.DepartmentName;
ERROR 1054 (42S22): Unknown column 'd.DepartmentName' in 'field list'
mysql> SELECT d.Dept_name,
    ->
    ->        COUNT(e.EmpID) AS TotalEmployees
    ->
    -> FROM Department d
    ->
    -> LEFT JOIN Emp e
    ->
    -> ON d.DeptID = e.DeptID
    ->
    -> GROUP BY d.DeptID, d.Dept_name;
+-----------+----------------+
| Dept_name | TotalEmployees |
+-----------+----------------+
| IT        |              3 |
| HR        |              2 |
| Sales     |              2 |
| Finance   |              0 |
+-----------+----------------+
4 rows in set (0.00 sec)

mysql> SELECT d.DeptName, COUNT(e.EmpID) AS TotalEmployees
    -> FROM Department d
    -> LEFT JOIN Emp e ON d.DeptID = e.DeptID
    -> GROUP BY d.DeptID, d.DeptName;
ERROR 1054 (42S22): Unknown column 'd.DeptName' in 'field list'
mysql> SELECT d.Dept_name, COUNT(e.EmpID) AS TotalEmployees
    ->
    -> FROM Department d
    ->
    -> LEFT JOIN Emp e ON d.DeptID = e.DeptID
    ->
```

MySQL 8.0 Command Line Cli

```
ERROR 1054 (42S22): Unknown column 'd.Deptname' in 'field list'
mysql> describe department;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| DeptID    | int         | NO   | PRI | NULL    |       |
| Dept_Name | varchar(50) | NO   |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)

mysql> SELECT d.DeptName,
    ->        COUNT(e.EmpID) AS TotalEmployees
    -> FROM Department d
    -> LEFT JOIN Emp e
    -> ON d.DeptID = e.DeptID
    -> GROUP BY d.DeptID, d.DeptName;
ERROR 1054 (42S22): Unknown column 'd.DeptName' in 'field list'
mysql> SELECT d.DeptName,
    ->
    ->        COUNT(e.EmpID) AS TotalEmployees
    ->
    -> FROM Department d
    ->
    -> LEFT JOIN Emp e
    ->
    -> ON d.DeptID = e.DeptID
    ->
    -> GROUP BY d.DeptID, d.Dept_name;
ERROR 1054 (42S22): Unknown column 'd.DeptName' in 'field list'
mysql> SELECT d.Dept_name,
    ->
    ->        COUNT(e.EmpID) AS TotalEmployees
    ->
    -> FROM Department d
    ->
    -> LEFT JOIN Emp e
    ->
    -> ON d.DeptID = e.DeptID
    ->
    -> GROUP BY d.DeptID, d.Dept_name;
+-----------+----------------+
| Dept_name | TotalEmployees |
+-----------+----------------+
| IT        |              3 |
| HR        |              2 |
| Sales     |              2 |
| Finance   |              0 |
```

MySQL 8.0 Command Line Cli

```
Query OK, 0 rows affected (0.01 sec)

mysql> select * from Emp where salary > 50000;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 80000.00 |     10 | 2023-01-25  |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   104 | Pranav  | 75000.00 |     10 | 2025-11-14  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
|   112 | Arjun   | 58000.00 |     10 | 2025-06-13  |
+-------+---------+----------+--------+-------------+
6 rows in set (0.01 sec)

mysql> update Emp set salary = salary * 1.10 where DeptID = 10;
Query OK, 3 rows affected (0.00 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> select * from Emp;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |
+-------+---------+----------+--------+-------------+
7 rows in set (0.00 sec)

mysql> select DeptID, count(*) as TotalEmployees from Emp Group by DeptID;
+--------+----------------+
| DeptID | TotalEmployees |
+--------+----------------+
|     10 |              3 |
|     20 |              2 |
|     30 |              2 |
+--------+----------------+
3 rows in set (0.01 sec)

mysql> select d.Deptname, count(e.EmpID) as TotalEmployees from Department d left join Emp e on d.DeptID = e.DptID group by d.DeptID, d.DeptName;
ERROR 1054 (42S22): Unknown column 'd.Deptname' in 'field list'
mysql> select d.Deptname, count(e.EmpID) as TotalEmployees from Department d left join Emp e on d.DeptID = e.DptID group by d.DeptID, d.Deptname;
ERROR 1054 (42S22): Unknown column 'd.Deptname' in 'field list'
```

MySQL 8.0 Command Line Cli

```
| GRANT SELECT ON `employee`.`emp` TO `readonly_user`@`localhost` |
+-----------------------------------------------------------------+
2 rows in set (0.00 sec)

mysql> Revoke insert on Emp From 'temp_user'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'temp_user' on host 'localhost' on table 'emp'
mysql> CREATE USER 'temp_user'@'localhost'
    -> IDENTIFIED BY 'Temp@123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT INSERT ON Emp TO 'temp_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE INSERT ON Emp FROM 'temp_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'temp_user'@'localhost';
+-----------------------------------------------+
| Grants for temp_user@localhost                |
+-----------------------------------------------+
| GRANT USAGE ON *.* TO `temp_user`@`localhost` |
+-----------------------------------------------+
1 row in set (0.00 sec)

mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into Emp values(112,"Arjun",58000,10,'2025-06-13');
Query OK, 1 row affected (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.02 sec)

mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> Update Emp set salary = 80000 where EmpID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> commit;
Query OK, 0 rows affected (0.01 sec)

mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> savepoint sp1;
```

MySQL 8.0 Command Line Cli

```
                                                
```

MySQL 8.0 Command Line Cli

```
mysql> update Emp set salary = 60000 where EmpID = 5;
Query OK, 0 rows affected (0.01 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> select * from Emp;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 60000.00 |     10 | 2023-01-25  |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   104 | Pranav  | 75000.00 |     10 | 2025-11-14  |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
+-------+---------+----------+--------+-------------+
6 rows in set (0.00 sec)

mysql> grant select on emp to 'readonly_user'@'localhost';
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> GRANt select on emp to 'readonly_user'@'localhost';
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> SELECT USER(), CURRENT_USER();
+----------------+----------------+
| USER()         | CURRENT_USER() |
+----------------+----------------+
| root@localhost | root@localhost |
+----------------+----------------+
1 row in set (0.00 sec)

mysql> SHOW GRANTS;
+-----------------------
```

MySQL 8.0 Command Line Cli

```
    ->     ->     -> (104,"Shruti",50000,30,'2024-05-16'),
    ->     ->     -> (105,"Om",90000,20,'2020-11-25'),
    ->     ->     -> (104,"Shruti",50000,30,'2024-05-16');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '-
>     -> (103,"Rani",75000,10,'2021-03-20'),
    ->     -> (104,"Shruti",50000,' at line 3
mysql> insert into Emp values (101,"Sakshi",60000,10,'2023-01-25'),(102,"Shruti",55000,20,'2023-11-12');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> insert into Emp values (103,"Sumit",70000,30,'2022-10-21'),(104,"Pranav",75000,10,'2025-11-14')(105,"Om",50000,20,'2020-10-11')(106,"Malhar",65000,30
,'2022-09-09');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(
105,"Om",50000,20,'2020-10-11')(106,"Malhar",65000,30,'2022-09-09')' at line 1
mysql> insert into Emp values (103,"Sumit",70000,30,'2022-10-21'),(104,"Pranav",75000,10,'2025-11-14')(105,"Om",50000,20,'2020-10-11'),(106,"Malhar",65000,3
0,'2022-09-09');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(
105,"Om",50000,20,'2020-10-11'),(106,"Malhar",65000,30,'2022-09-09')' at line 1
mysql> insert into Emp values (103,"Sumit",70000,30,'2022-10-21'),(104,"Pranav",75000,10,'2025-11-14'),(105,"Om",50000,20,'2020-10-11'),(106,"Malhar",65000,
30,'2022-09-09');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> desc Emp;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| EmpID       | int           | NO   | PRI | NULL    |       |
| EmpName     | varchar(50)   | NO   |     | NULL    |       |
| Salary      | decimal(10,2) | YES  |     | NULL    |       |
| DeptID      | int           | YES  | MUL | NULL    |       |
| JoiningDate | date          | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> select * from Emp;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 60000.00 |     10 | 2023-01-25  |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   104 | Pranav  | 75000.00 |     10 | 2025-11-14  |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
+-------+---------+----------+--------+-------------+
6 rows in set (0.00 sec)
```

MySQL 8.0 Command Line Cli

```
mysql> Insert into Department values(10,"IT"),(20,"HR"),(30, "Sales"),(40 ,"Finance");
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from department;
+--------+-----------+
| DeptID | Dept_Name |
+--------+-----------+
|     10 | IT        |
|     20 | HR        |
|     30 | Sales     |
|     40 | Finance   |
+--------+-----------+
4 rows in set (0.00 sec)

mysql> insert into Emp(101,"Sakshi", 60000,10,'2023-01-25'),
    -> (102,"Pranav",45000,20,'2022-09-15'),
    -> (103,"Rani",75000,10,'2021-03-20'),
    -> (104,"Shruti",50000,30,'2024-05-16'),
    -> (105,"Om",90000,20,'2020-11-25'),
    -> (106,"Sumit",55000,30,'2023-08-12');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '1
01,"Sakshi", 60000,10,'2023-01-25'),
(102,"Pranav",45000,20,'2022-09-15'),
(103' at line 1
mysql>  insert into Emp(101,"Sakshi", 60000,10,'2023-01-25'),
    ->      -> (102,"Pranav",45000,20,'2022-09-15'),
    ->      -> (103,"Rani",75000,10,'2021-03-20'),
    ->      -> (104,"Shruti",50000,30,'2024-05-16'),
    ->      -> (105,"Om",90000,20,'2020-11-25'),
    ->      -> (105,"Om",90000,20,'2020-11-25'),^Z
    -> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '1
01,"Sakshi", 60000,10,'2023-01-25'),
    -> (102,"Pranav",45000,20,'2022-09-15'' at line 1
mysql>  insert into Emp(101,"Sakshi", 60000,10,'2023-01-25'),
    ->      -> (102,"Pranav",45000,20,'2022-09-15'),
    ->      -> (103,"Rani",75000,10,'2021-03-20'),
    ->      -> (104,"Shruti",50000,30,'2024-05-16'),
    ->      -> (105,"Om",90000,20,'2020-11-25'),
    ->      -> (104,"Shruti",50000,30,'2024-05-16');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '1
01,"Sakshi", 60000,10,'2023-01-25'),
    -> (102,"Pranav",45000,20,'2022-09-15'' at line 1
mysql> insert into Emp values (101,"Sakshi",60000,10,'2023-01-25'),
    -> (102,"Pranav",45000,20,'2022-09-15'),
    ->      ->      -> (103,"Rani",75000,10,'2021-03-20'),
```

MySQL 8.0 Command Line Cli

```
Enter password: *************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database Employee;
Query OK, 1 row affected (0.01 sec)

mysql> use Employee;
Database changed
mysql> create table Department(DeptID int primary key, Dept_Name varchar(50) not null);
Query OK, 0 rows affected (0.03 sec)

mysql> desc Department;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| DeptID    | int         | NO   | PRI | NULL    |       |
| Dept_Name | varchar(50) | NO   |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
2 rows in set (0.02 sec)

mysql> create table Emp(EmpID int primary key, EmpName varchar(50) not null, Salary decimal(10,2), DeptID int, JoiningDate date, foreign key(DeptID) referen
ces Department(DeptID));
Query OK, 0 rows affected (0.03 sec)

mysql> desc Emp;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| EmpID       | int           | NO   | PRI | NULL    |       |
| EmpName     | varchar(50)   | NO   |     | NULL    |       |
| Salary      | decimal(10,2) | YES  |     | NULL    |       |
| DeptID      | int           | YES  | MUL | NULL    |       |
| JoiningDate | date          | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> Insert into Department values(10,"IT"),(20,"HR"),(30, "Sales"),(40 ,"Finance");
```